

Fundamentals of Information Systems Security

Nmap Network Scanning Cheat Sheet

Version: 1.0

The Network Mapper

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1 Basic Scan Commands

Command	Syntax	Description
Ping Scan	<code>nmap -sn 192.168.1.0/24</code>	Discover live hosts (no port scan)
Quick Scan	<code>nmap -F 192.168.1.1</code>	Scan top 100 ports
Standard Scan	<code>nmap 192.168.1.1</code>	Default SYN scan of top 1000 ports
Full Scan	<code>nmap -p- 192.168.1.1</code>	Scan all 65,535 ports (slow)
Specific Ports	<code>nmap -p 80,443 192.168.1.1</code>	Scan specific ports
Port Range	<code>nmap -p 1-1000 192.168.1.1</code>	Scan port range
Service Version	<code>nmap -sV 192.168.1.1</code>	Detect service versions
OS Detection	<code>nmap -O 192.168.1.1</code>	Detect operating system
Aggressive Scan	<code>nmap -A 192.168.1.1</code>	OS, version, scripts, traceroute

2 Scan Types

Scan Type	Syntax	Description
TCP SYN Scan	<code>nmap -sS 192.168.1.1</code>	Stealth, default (needs root)
TCP Connect Scan	<code>nmap -sT 192.168.1.1</code>	Full TCP connection (no root)
UDP Scan	<code>nmap -sU 192.168.1.1</code>	Scan UDP ports (slow)
TCP Null Scan	<code>nmap -sN 192.168.1.1</code>	Null flags (evasive)
TCP FIN Scan	<code>nmap -sF 192.168.1.1</code>	FIN flags (evasive)
TCP Xmas Scan	<code>nmap -sX 192.168.1.1</code>	FIN, PSH, URG flags (evasive)
ACK Scan	<code>nmap -sA 192.168.1.1</code>	ACK flags (firewall mapping)
Window Scan	<code>nmap -sW 192.168.1.1</code>	TCP window scan
Idle Scan	<code>nmap -sI zombie 192.168.1.1</code>	Zombie scan (true stealth)
Ping Sweep	<code>nmap -sP 192.168.1.0/24</code>	ICMP echo requests

3 Important Scan Options

Option	Syntax	Description
Service Version	<code>-sV</code>	Probe open ports for service/version info
Version Intensity	<code>-sV --version-intensity 5</code>	Set version probe intensity (0-9)
OS Detection	<code>-O</code>	Enable OS detection
Aggressive Detection	<code>-A</code>	Enable OS, version, script, and traceroute
Trace Route	<code>--traceroute</code>	Trace path to host

Script Engine	-script=script	Run NSE scripts
Verbose Output	-v	Increase verbosity
Very Verbose	-vv	Very verbose output
Debugging	-d	Debug output
Packet Timing	-T4	Timing template (0-5)
Randomize Hosts	-randomize-hosts	Randomize target order
No DNS	-n	Skip DNS resolution
DNS Resolution	-R	Always resolve DNS

4 Nmap Scripting Engine (NSE)

NSE Script Categories

- **auth** - Authentication bypass
- **broadcast** - Broadcast discovery
- **brute** - Brute force attacks
- **default** - Default scripts (-sC)
- **discovery** - Service discovery
- **dos** - Denial of service
- **exploit** - Exploitation
- **external** - External resources
- **fuzzer** - Fuzzing
- **intrusive** - Intrusive scans
- **malware** - Malware detection
- **safe** - Safe scripts
- **version** - Version detection
- **vuln** - Vulnerability detection

4.1 Common NSE Scripts

Script	Usage	Description
http-headers	nmap -script=http-headers target	HTTP headers
http-enum	nmap -script=http-enum target	Directory enumeration
smb-os-discovery	nmap -script=smb-os-discovery target	SMB OS discovery
smb-vuln-*	nmap -script=smb-vuln-* target	SMB vulnerabilities
ftp-anon	nmap -script=ftp-anon target	Anonymous FTP login
ssh-brute	nmap -script=ssh-brute target	SSH brute force
mysql-info	nmap -script=mysql-info target	MySQL info
snmp-info	nmap -script=snmp-info target	SNMP info

rdp-enum-encryption	nmap --script=rdp-enum-encryption target	RDP encryption
dns-recursion	nmap --script=dns-recursion target	DNS recursion

5 Output Options

Option	Syntax	Description
Normal Output	-oN output.txt	Normal output to file
XML Output	-oX output.xml	XML output to file
Grepable Output	-oG output.txt	Grepable format
All Formats	-oA output	All formats (N, X, G)
Script Output	--script-args	Pass arguments to scripts

6 Timing Templates

Template	Flag	Description
Paranoid	-T0	Very slow, avoids IDS detection
Sneaky	-T1	Slow, evasive scanning
Polite	-T2	Slower, less bandwidth
Normal	-T3	Default timing
Aggressive	-T4	Fast scanning
Insane	-T5	Very fast, may drop packets

7 Common Scan Examples

Real-World Scan Examples

```
# Discover live hosts on network
nmap -sn 192.168.1.0/24

# Quick port scan with version detection
nmap -sV -F 192.168.1.1

# Full port scan with OS detection
nmap -p- -O 192.168.1.1

# Aggressive scan with scripts
nmap -A -sC 192.168.1.1

# Scan UDP ports 53,67,68,161
nmap -sU -p 53,67,68,161 192.168.1.1

# Scan with specific NSE script
nmap --script=smb-vuln-* 192.168.1.1

# Save output to all formats
nmap -A -oA scan_results 192.168.1.1

# Scan a range of IPs
nmap 192.168.1.1-254

# Scan subnet with service detection
nmap -sV 192.168.1.0/24
```

8 Port States

Port State Definitions	
State	Description
Open	Port is open and accepting connections
Closed	Port is closed (no service listening)
Filtered	Port is filtered (firewall/protection)
Unfiltered	Port is reachable but state unknown
Open Filtered	Open or filtered (no response)
Closed Filtered	Closed or filtered (no response)